

The School of Mathematics



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Density-Aware Graph Geodesic Stress Testing on a Learned Macroeconomic Manifold

by

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Dissertation Presented for the Degree of
MSc in Computational Applied Mathematics

August 2026

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Abstract

[abstract]

Acknowledgments

[acknowledgments]

Own Work Declaration

[own work declaration]

AI disclosure

[see guidance document]

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1 Introduction

What is the problem, why do TNP care, how are you going to solve it? **What do we intend to add?**

Let $z_t \in \mathbb{R}^{m+n}$, $t = 1, \dots, N$, denote a transformed and standardised macroeconomic state vector. The aim is to learn a low-dimensional representation $\psi_t \in \mathbb{R}^\ell$ using diffusion maps or a closely related anisotropic diffusion-map construction, and then to construct stress scenarios as paths on the learned state manifold. The baseline stress path between a benign state and a stress state is usually linear:

$$\gamma_{\text{lin}}(s) = (1 - s)\psi_{\text{start}} + s\psi_{\text{end}}, s \in [0, 1].$$

The dissertation replaces this with graph-based stress paths that follow the learned data geometry. The central path is a density-aware graph shortest path:

$$\gamma_\beta = \arg \min_{\Gamma: i_{\text{start}} \rightarrow i_{\text{stress}}} \sum_{(i,j) \in \Gamma} \|\psi_i - \psi_j\| \exp \left(\beta \frac{V_i + V_j}{2} \right)$$

, where $V_i = -\log(\max(\hat{\rho}_i, \delta))$. Here $\hat{\rho}_i$ is an empirical density proxy at ψ_i , $\delta > 0$ is a density floor, and $\beta \geq 0$ controls the penalty for travelling through historically sparse regions.

2 Background

In this section we introduce the foundational mathematical concepts from which we develop our method of scenario generation.

2.1 State vectors and data manifolds

A macroeconomic state vector is a point

$$z_t = (z_t^1, \dots, z_t^D) \in \mathbb{R}^D$$

representing a variety of different macroeconomic variables at a time t , e.g. fuel prices, interest rates, etc. Although D may be large, the collection of observed states may concentrate near a lower-dimensional set

$$\mathcal{M} \in \mathbb{R}^D.$$

This is the manifold hypothesis. In this dissertation, the manifold is not assumed known analytically; instead, it is estimated numerically from data.

2.2 Euclidean distances and Gaussian kernels

In order to build our diffusion map, we need a notion of affinities between points in our data cloud. To this end, we find the squared Euclidean distance between two standardised macroeconomic states as

$$D_{ij} = \|z_i - z_j\|^2,$$

and we calculate an affinity matrix using a Gaussian kernel:

$$W_{ij} = \exp\left(-\frac{D_{ij}}{\varepsilon}\right).$$

The bandwidth ε controls locality: a small ε means the affinity matrix becomes nearly diagonal and the points become disconnected; a large ε means that the kernel saturates and the graph loses geometric resolution.

We can remove the effect of density via α -normalisation:

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^\alpha q_j^\alpha},$$

where $q_i = \sum_j W_{ij}$.

2.3 Diffusion maps

Diffusion maps allow us to map our high-dimensional state vector to a lower-dimensional space. We compute a Markov matrix

$$P_{ij} = \frac{W_{ij}^{(\alpha)}}{\sum_k W_{ik}^{(\alpha)}},$$

and the diffusion coordinates are obtained from the leading non-trivial eigenvectors of P :

$$\psi_i = (\lambda_1 \phi_1(i), \dots, \lambda_r \phi_r(i)) \in \mathbb{R}^r.$$

These coordinates are designed to preserve diffusion geometry: two points are close if random walks starting from them behave similarly.

2.4 Kernel mass and scale selection

As mentioned above, choosing the correct bandwidth is a very important step in the process of finding our diffusion map. For a clean d -dimensional manifold, we have the asymptotic relation $T(\varepsilon) \sim C\varepsilon^{d/2}$, where

$$T(\varepsilon) = \sum_{i,j} W_{ij}(\varepsilon).$$

Then $\log T(\varepsilon) \approx \frac{d}{2} \log \varepsilon + \log C$, and therefore

$$\frac{d \log T(\varepsilon)}{d \log \varepsilon} \approx \frac{d}{2}.$$

If the intrinsic dimension d is known, this relation can help us find an appropriate bandwidth ε . In the case of noisy macroeconomic data, this method is only diagnostic and should be combined with spectral analysis and downstream path robustness.

2.5 Graphs and shortest paths

Once we have found our diffusion map coordinate system, we need a way to traverse this space. For this we create a k -nearest-neighbour graph from the coordinates $\psi_i \in \mathbb{R}^r$. A graph consists of nodes and edges:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}),$$

where in our case nodes represent macroeconomic variables in diffusion-coordinate space and edges connect each node to its k nearest neighbours. A path is a sequence of nodes $\Gamma = (i_0, i_1, \dots, i_L)$ such that each adjacent pair is connected by an edge.

Given edge weights w_{ij} , the cost of a path is

$$C(\Gamma) = \sum_{\ell=0}^L w_{i_\ell i_{\ell+1}}.$$

The shortest path between nodes is the path with the minimal cost.

2.6 Unknown latent process

Explain the model assumptions and how the graph Laplacian approximates the generator of the hidden diffusion process.

3 Numerical models

4 Data

4.1 Data gathering and auditing

For our $x(t)$ series, I have chosen to use a subset of the Federal Reserve Economic Data monthly dataset (FRED-MD). The subset was chosen to represent a wide range of the economy covering output and activity levels, employment, inflation, interest rates, credit spreads, liquidity, and financial market stress proxies.

For the price series $y(t)$, we choose to use the “10 Industry Portfolios” dataset available from the Noreen R. French website. This dataset tracks the monthly returns of 10 different portfolios. The portfolios are split into the following industries: consumer non-durables (such as food and tobacco); consumer durables (such as cars and TVs); manufacturing; energy; tech; telecommunications; shops; health; utilities; and other sectors (such as mines, hotels, and finance). The portfolios are updated yearly to include all eligible companies within each industry. This dataset was chosen to represent the general asset trends in an economy. Choosing individual asset prices going back to 1963 would mean we run into survivorship bias, where we would necessarily only include companies which have managed to stay afloat for the entire 63-year period. These companies do not represent the majority of companies in the economy.

4.2 Data preprocessing

Extensive data auditing was completed; flatlines were checked and missing values were investigated. Out of 45 variables across 755 months, there were 4 missing values due to a U.S. government shutdown in October 2025. These values were linearly interpolated.

Many of the time series in the FRED-MD dataset had extreme outliers, which caused problems for the diffusion coordinate calculations. We therefore had to spend a lot of time finding a way to sensibly mitigate the impact of these outliers.

Transformations were applied to the time series to ensure stationarity. FRED-MD documentation supplied us with the recommended month-on-month (MoM) transforms which were applied. However, for a selection of activity variables, we opted to use year-on-year (YoY) transforms. This helped dampen the impact of crisis events like COVID-19, while keeping economic interpretability. **Include graph comparing MoM to YoY?**

Next, several standardisation methods were compared. A general standardisation function $S(Z)$ acts on the transformed observation matrix and returns a standardised matrix

$$\tilde{Z} = \begin{bmatrix} \tilde{z}_1^\top \\ \vdots \\ \tilde{z}_N^\top \end{bmatrix} \in \mathbb{R}^{Nx D}.$$

Initially, we used a basic approach where each coordinate was standardised via

$$S(Z)_{tj} = \frac{z_t^j - \bar{z}_t}{s_j}, \quad j = 1, \dots, D,$$

where s_j is the standard deviation of row j . However, when computing the diffusion coordinates with this data, we encountered a problem. The first several diffusion coordinates would only be able to identify the location of one of the outliers in the data; it would be completely flat elsewhere. (Looks like the dirac delta function)

After comparing several different standardisation methods, we decided to choose the quantile method. **DESCRIBE THIS METHOD - RAISE CONCERNS - JUSTIFY THE USE OF IT HERE.** Groups value into quantiles, keeps ordering, does not keep scales!

FIGURE OF A SINGLE DIFFUSION COORDINATE SHOWING HOW DIFFERENT STANDARDISATIONS LEAD TO DIFFERENT TIME SERIES - Extreme point indicator vs. Noisy plot

5 Numerical methods

5.1 Bandwidth selection

As mentioned in 2 investigating the kernel mass plots does not give us enough information on how to choose the bandwidth. Instead of solely relying on this tool, we also investigate the spectrum of the computed transition matrix P and the degrees of the adjacency matrix W . We seek a bandwidth which gives us graph where no points are disconnected, i.e. the minimum degree must be greater than 1 and the coefficient of variance of the degrees must be stable. Furthermore, we aim to find a bandwidth that gives us a clear and large spectral gap for the transition matrix. This implies that the embedding is in fact picking up some underlying structure.

PLOT OF SPECTRUM??

After choosing the appropriate bandwidth we choose our embedding dimension, signified by the position of the largest gap in eigenvalues. For our data we find an embedding dimension of $r = 3$ is appropriate. In Figure 1 we justify our choice of embedding dimension by highlighting that some variables are impossible to describe in only 2 dimenions but are cleanly described in 3.

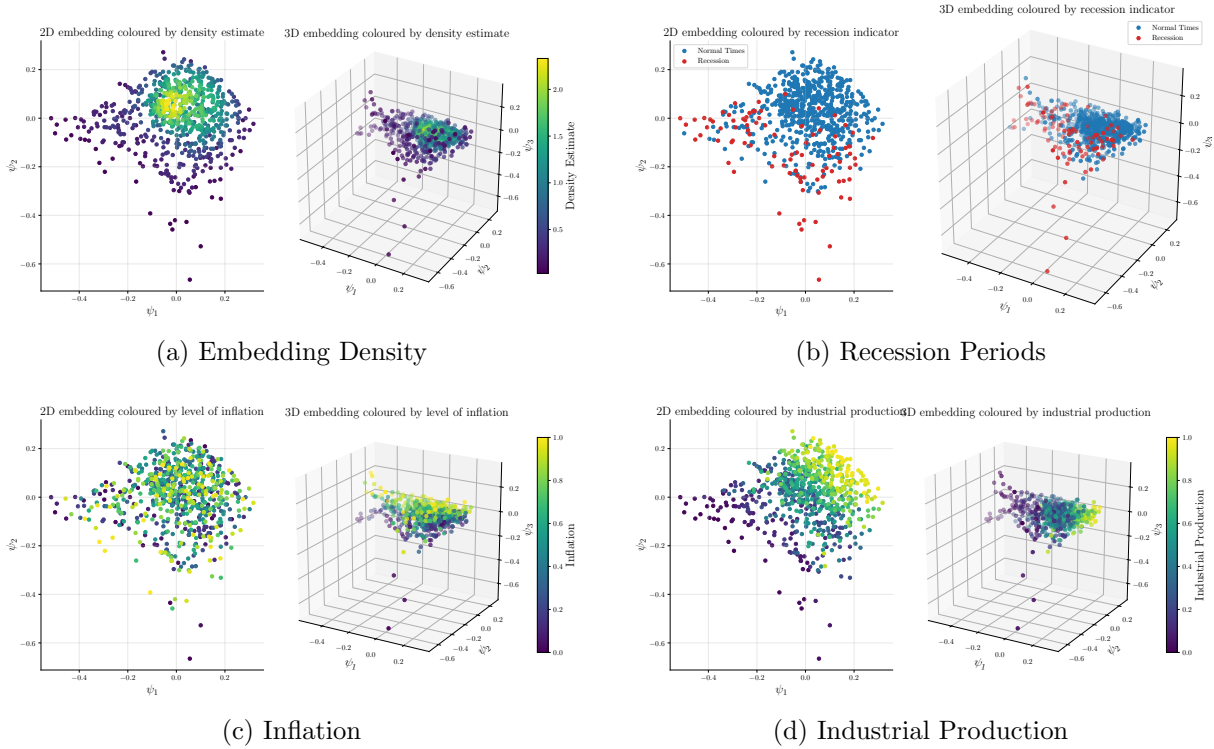


Figure 1: 2D and 3D Latent Embeddings comparison across different economic indicators.

TITLES AND LABELS ARE TOO SMALL. DO I INCLUDE ALL OF THESE? HOW DO I FORMAT THEM SO I CAN READ THE LABELS?

5.2 Validation on toy model

Maybe put the starter stuff here with the OU processes

5.3 Density estimate of the diffusion embedding

Now that we have our diffusion embedding, we focus our work on how we traverse this space. With the diffusion coordinates, we create a k -nearest-neighbours graph which connects node i to its k nearest neighbours. We the symmetrise this graph by taking the union of directed nearest-neighbour edges, this means that now each node is connected to potentially more than k other nodes. We construct graphs for multiple values of k and choose whether to accept or

reject the resulting graph by ensuring a graph passes a series of tests. Namely, a graph passes if it is connected; it's not nearly complete (a complete graph is one where all of the nodes are connected to each other); typical node edge is reasonable; the shortest paths between nodes in the graph are not dominated by a few very large jumps; and the results must be stable for nearby k . For our diffusion coordinates, all values of $k \in \{5, 10, 15, 20, 25, 30\}$ passed our tests, so we chose a value $k = 15$ to continue.

For a graph with weights

$$w_{ij}^{(0)} = \|\psi_i - \psi_j\|,$$

the shortest path approximates a geodesic path through the empirical cloud. It is not a straight chord through empty space; it must move through observed neighbouring states. We now aim to develop a method which controls the density of points we visit along our path. We must, therefore, build a notion of density in the diffusion-coordinate space. For that, we use the diffusion-map kernel degrees:

$$\hat{\rho}_i \propto q_i = \sum_{j=1}^N \exp\left(-\frac{\|\psi_i - \psi_j\|^2}{h}\right),$$

where h is a bandwidth parameter chosen to be one tenth of the median squared distance between the diffusion coordinates. The proportionality constant can be ignored because we will work with the relative density. We then apply a density floor: $\hat{\rho}_{i,\delta} = \max\{\hat{\rho}_i, \delta\}$ and we define the density potential $V_i = -\log \hat{\rho}_{i,\delta}$.

Then we define the density-aware edge cost as

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right),$$

thus making the 'shortest path' depend on the density of the region it passes through. Increasing β means that passing through low-density regions becomes more expensive; therefore, the shortest path avoids these regions.

5.4 Choice of economic endpoints

We aim to generate paths from benign economic states to stressed economic states. A hypothetical shocked macro vector z_{shock} is not automatically on the learned manifold and does not automatically have a diffusion coordinate ψ_{shock} . Mapping it into the learned manifold is an out-of-sample inverse problem. To keep this dissertation focused, historical endpoints are used as the primary stress endpoints. We choose sets of two observed dates: t_{start} and t_{end} , then

$$\psi_{\text{start}} = \psi(t_{\text{start}}), \quad \psi_{\text{end}} = \psi(t_{\text{end}}).$$

Table 1 shows our choice of endpoints.

Table 1: Macroeconomic Regimes and Event Analysis

Regime / Macro Event	Start Date	End Date
Benign Expansion to Global Financial Crisis (GFC)	2006-03-01	2008-10-01
Benign Expansion to COVID-19 Shock	2019-07-01	2020-04-01
Low-Inflation to Monetary Tightening	2019-04-01	2022-04-01
Pre-Crisis to Early 1980s Recession Trough	1977-01-01	1982-07-01

Each of the endpoints corresponds to a specific point in time where the economy was in a certain state. Figure 2 shows 4 different paths between each of the endpoints in diffusion-coordinate space.

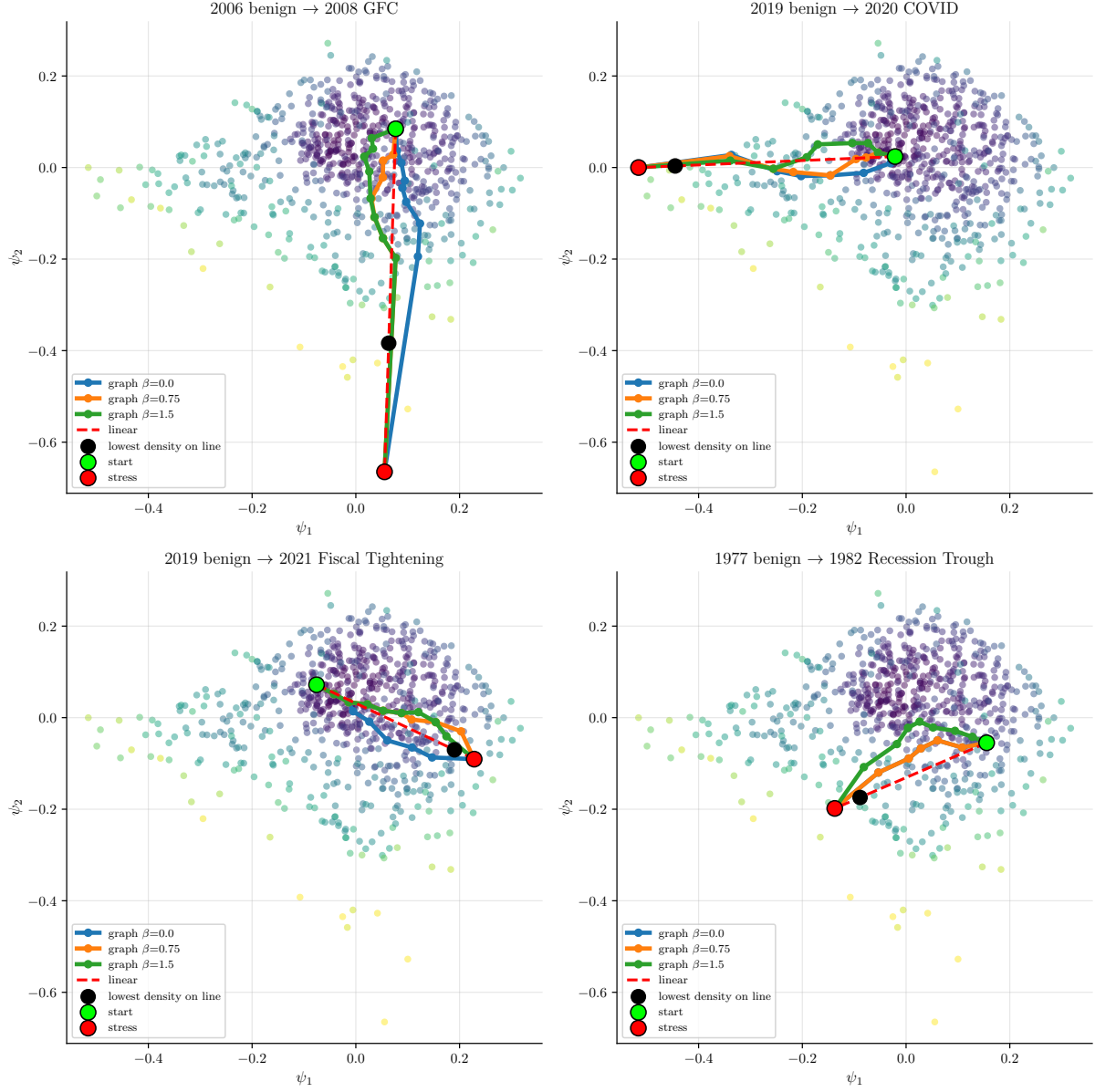


Figure 2: Paths between benign and stressed scenarios in diffusion-coordinate space

5.5 Lifting method

In this dissertation, we compare the density of paths in the diffusion-coordinate space. We compare our graph paths with a linear path. While our main result comes from the densities in diffusion space, we also would like to compare the paths in the full macroeconomic space. To do this we need to have a way to lift our low dimensional paths in diffusion space into the macro space. In [1] they used a linear operator H in order to project back into the macro space by

$$\tilde{z}_t \approx H\psi_t.$$

In practice, H is approximated by

$$H = \arg \min_{B \in \mathbb{R}^{N \times D}} \sum_{i=1}^N \|\tilde{z}_t - B\psi_t\|^2.$$

In this dissertation, we use a more nuanced approach by utilising our k -nearest-neighbours graph in a method called local-neighbourhood lifting (LNL). For any path in diffusion space $\gamma(s)$, we define our lifted coordinated $\hat{z}(s)$ as

$$\hat{z}(s) = \sum_{j \in \mathcal{N}_m(\gamma(s))} a_j(s) z_j,$$

where $\mathcal{N}_m(\gamma(s))$ is a set of the m nearest points to $\gamma(s)$. and weights $a_j(s)$ satisfy

$$a_j(s) \geq 0, \quad \sum_j a_j(s) = 1.$$

We choose to use Gaussian weights:

$$a_j(s) = \frac{\exp(-\|\gamma(s) - \psi_j\|^2/\tau)}{\sum_{k \in \mathcal{N}_m(\gamma(s))} \exp(-\|\gamma(s) - \psi_k\|^2/\tau)}.$$

To validate the LNL method and compare it with the Baker et al. style method, we lifted linear paths between our chosen endpoints. We then calculate the distances from each point on the projected path to the nearest training observation in macro space. Formally, $d_{NN}(\hat{z}_\ell) = \min_{1 \leq i \leq N} \|\hat{z}_\ell - \tilde{z}_i\|$. We compare these distances with those between the training points in macro space. Specifically, we compare the 95th-percentile of the training set nearest-neighbour distances to that of the lifted path set. For all paths, the LNL method worked better than the Baker-style method. In all cases, the LNL method managed to keep its 95-percentile distance below that of the ambient space, suggesting that all of the lifted coordiantes were not wildly away from the observed data cloud. The Baker-style lift, on the other hand, failed this test in the 2008 GFC path, implying that this lifting method may not be suitable in all stress scenarios.

Is there more to say about this? This is sort of suggesting that linear paths and LNL are realistic, right? The linear paths are in low density regions, but the lifted paths with LNL don't seem to raise any red flags. Clearly the graph paths have a d_{NN} of zero, but this is not great for generating new scenarios. Should I talk about this here?

6 Numerical experiments

6.1 Minimum density results

The main aim of this dissertation is to show that traversing along density-aware paths in latent space gives more plausible price paths than strictly linearly interpolating between two points. We aim to show that

$$M_{\min}(\gamma) < M_{\min}(\Gamma_{\beta}),$$

where $M_{\min}(\gamma) = \min \hat{\rho}(\gamma(s))$ and $M_{\min}(\Gamma_{\beta}) = \min \hat{\rho}_i$. To compute the density of a point on the line in latent space, which does not correspond to a training point, we use the diffusion-map kernel degree formula

$$\hat{\rho}(\gamma_i) \sum_{j=1}^N \exp \left(-\frac{\|\gamma_i - \psi_j\|^2}{h} \right),$$

using the same bandwidth parameter used to calculate the densities of the training points.

In Table 2 we compare the minimum densities of different paths in latent space between our four chosen endpoint pairs. For every event, and every level of β we see that $M_{\min}(\gamma) <$

Table 2: Minimum Linear Density ($\min \rho_{\text{lin}}$) Across Transition Paths

Transition Event	Linear	Graph Path		
		$\beta = 0.0$	$\beta = 0.75$	$\beta = 1.5$
2006 Benign $\rightarrow$ 2008 GFC	0.0031	0.3248	0.4086	0.4086
2019 Benign $\rightarrow$ 2020 COVID	0.0857	0.3584	0.3584	0.4019
2019 Benign $\rightarrow$ 2021 Fiscal Tightening	0.8409	1.0984	1.1568	1.1568
1977 Benign $\rightarrow$ 1982 Recession Trough	0.2410	0.3830	0.3444	0.5641

$M_{\min}(\Gamma_{\beta})$. Further, we see that increasing the value of β increases the value of the minimum density along a path. **TALK ABOUT HOW MEAN DENSITY DID NOT NECESSARILY INCREASE AND EXPLAIN WHY.**

6.2 Lifting method results

I also have these results but need to figure out what to put in each section... Right now I have put all of this in the methods section, I have just described the outcomes.

6.3 Inverse standardisation...

I have not got results for this. I don't know if I will even do this. There are more interesting things we can do in the standardised space before we try to figure out how to come back.

6.4 Robustness checks

We check for the robustness of our method by varying two key variables, the strength of sparsity avoidance β and the number of neighbours used to construct the diffusion-coordinate graph k . We expect to see the results of our test vary smoothly when changing parameters.

In Figure 3 we see the minimum density of graph paths smoothly increasing as β increases. The smoothness indicates robustness and the upward path makes sense as the path is being penalised more and more for going to low density regions, therefore the minimum density increases.

In Figure 4 we see that again, all of the metrics change smoothly across different levels of k . Furthermore, the direction of change can be easily understood. The mean interior density metric is monotonically increasing for each endpoint pair. By increasing k , we are giving each node more neighbours, i.e. more options to choose the next step. Since the shortest path minimises

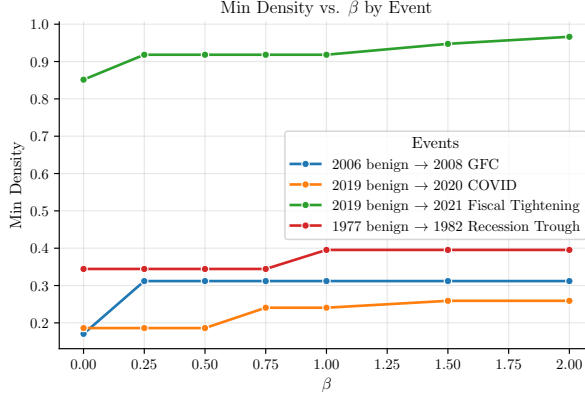


Figure 3: Effect of increasing β on the minimum density of a density-aware path

the minimum density, then adding new points cannot ever increase the minimum density, as a path including one of these new neighbours would not be chosen. Similarly the density weighted cost decreases as k increases. The path length metric decreases as the number of neighbours increases. This reflects the distance still being a factor in the cost, if the path can choose between two nodes of a very similar density but one is closer than the other, then it is going to choose the closest one. The roughness of the paths also decreases as k increases, this can be understood similarly to the path length. The only path where there is not a very obvious decrease in the direction change metric is the 1977 $\rightarrow$ 1982 Recession path, where the small- k paths are forced to travel along straight paths to the stress point, whereas the paths with lots of available neighbours were able to detour through a very dense region, thus reducing the cost.

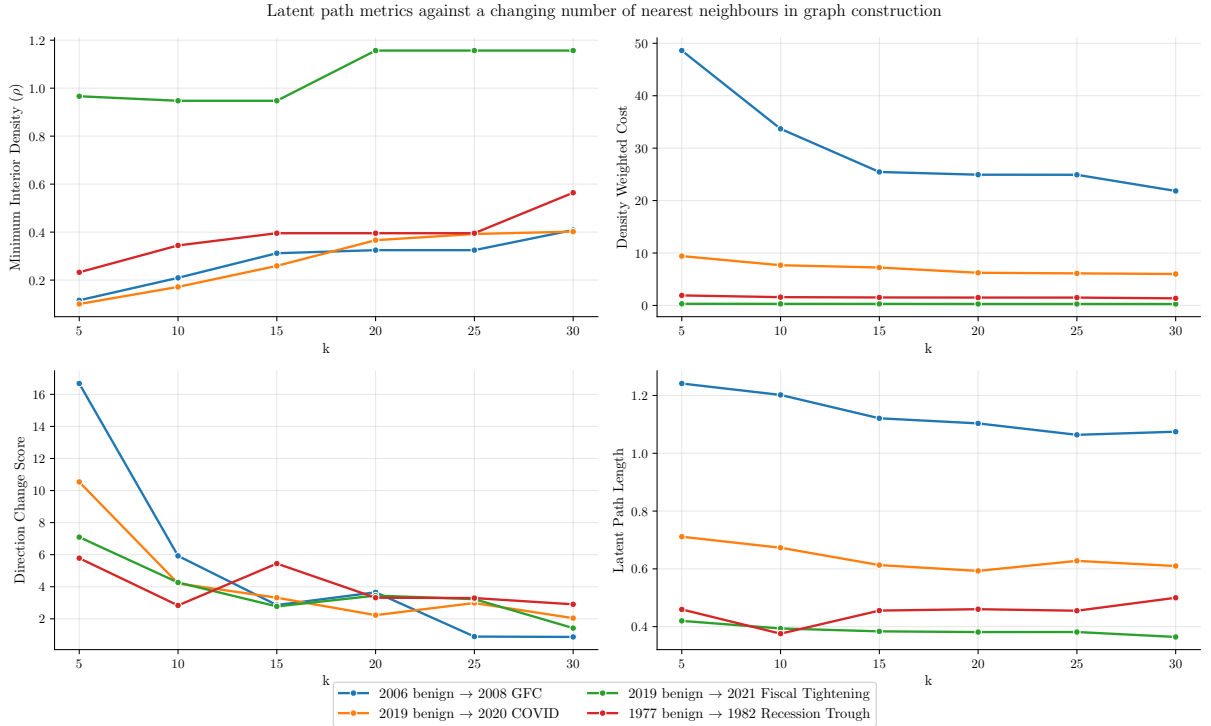


Figure 4: Varying the number of nearest neighbours in embedding graph construction's effect on several metrics

7 Conclusions and outlook

What does the JDKF paper do?

The problem is that we have high-dimensional, nonlinear time-series data, and we want to model the dynamical system which governs the series. Specifically, given key financial data including interest rates, inflation rates, employment, GDP growth, etc. and the prices of various assets, we want to be able to derive the dynamical system which drives the movement of these data points, allowing us to predict/generate price data for assets in an economy. A key idea of this project is being able to generate new scenarios which are economically feasible.

In our setup, we assume that we have a high-dimensional set of covariates $x(t) \in \mathbb{R}^m$ that drives a set of response variables $y(t) \in \mathbb{R}^n$. The covariates $x(t)$ represent the macroeconomic factors that drive the prices of financial assets in an economy $y(t)$. We assume that $m \gg n$, signifying that a few financial and macroeconomic variables drives all the prices in the economy; this is the main assumption of the factor model (Is it/maybe other models/citation?).

To work with high-dimensional data, we must employ a dimensionality reduction scheme. This scheme must be able to capture the nonlinearity of the original data. In [1], they suggested learning embeddings from $z(t) = (x(t), y(t)) \in \mathbb{R}^{m+n}$, rather than just $x(t)$, which is more typical is it more typical? Cite. Learning embeddings this way is motivated by the idea of keeping the model's results economically plausible, since embedding $x(t)$ alone may discard relationships between $x(t)$ and $y(t)$.

As in [1], we assume that the covariates and responses are driven by an unseen, lower-dimensional system $\theta_t \in \mathbb{R}^d$ satisfying a Langevin diffusion via nonlinear functions. These nonlinear functions restrict the trajectories $z(t)$ to some unknown manifold $\mathcal{M}$.

For concreteness, we consider an $\mathbb{R}^d$ -valued process $\theta = (\theta)_{t \geq 0}$ and an $\mathbb{R}^{m+n}$ -valued process $Z = (Z)_{t \geq 0}$ satisfying

$$d\theta_t = -\nabla U(\theta_t)dt + \sqrt{2}dW_t, Z_t = H(\theta_t) \quad (7.1)$$

where W is d -dimensional Brownian motion. The potential $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is assumed to be such that the infinitesimal generator of θ , given by $\mathcal{L} := \Delta - \nabla U \cdot \nabla$, possesses a discrete spectrum.

Given $N + 1$ discrete observations $\{Z(t_i)\}_{i=0}^N$, and without knowledge of U , H , or the latent dimension d , we wish to model the process Z in order to make predictions, generate scenarios, etc.

In [1], they adopt a spectral approach, whereby H is approximated using $\ell \geq 1$ principal eigenfunctions $(\varphi_1, \dots, \varphi_\ell)$ of the generator $\mathcal{L}$. We have that $Z_t = H(\theta_t) \approx \mathbf{H}(\varphi_1(\theta_t), \dots, \varphi_\ell(\theta_t))^\top$ for some $\mathbf{H} \in \mathbb{R}^{(m+n) \times \ell}$. Using Itô's formula, if φ_i is associated with eigenvalue $-\lambda_i < 0$, then

$$d\varphi_i = -\lambda_i \varphi_i(\theta_t)dt + \sqrt{2} \nabla \varphi_i(\theta_t) \cdot dW_t \quad (7.2)$$

Finally, we can approximate $(\varphi_1(\theta_t), \dots, \varphi_\ell(\theta_t))^\top$ by a linear diffusion $\xi = (\xi_t)_{t \geq 0} \in \mathbb{R}^\ell$ and so we arrive at a *linear* hidden Markov Model

$$d\xi_t = -\Lambda \xi_t dt + \Gamma dB_t, Z_t = \mathbf{H} \xi_t + \eta_t, \quad (7.3)$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_\ell)$, $\Gamma \in \mathbb{R}^\ell$, and B is a standard ℓ -dimensional Brownian motion.

For us, solving this problem amounts to estimating Λ and $\mathbf{H}$.

The key insight of the JDKF is that the leading eigenfunctions of the latent generator need not be computed directly. Instead, they can be approximated from the observed data using diffusion maps. Given the observations $\{Z(t_i)\}_{i=0}^N$, a kernel-based graph is constructed whose leading eigenvectors ψ_i converge, under suitable sampling assumptions, to the leading eigenfunctions φ_i of the generator $\mathcal{L}$. Consequently, the diffusion-map coordinates $\Psi(Z_t) = (\psi_1(Z_t), \dots, \psi_\ell(Z_t))$ provide a data-driven approximation of the intrinsic coordinates $(\varphi_1(\theta_t), \dots, \varphi_\ell(\theta_t))$. The JDKF therefore operates entirely on observable data while retaining information about the latent dynamics.

References

- [1] G. Baker, A. Capponi, and J. A. Sidaoui. Data-driven dynamic factor modeling via manifold learning. *arXiv preprint arXiv:2506.19945*, 2025.

Appendices

A First appendix

B Second appendix